

Task 7

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1) Cloudflare — 7.3 Tbps attack (mid-May 2025)

Target: a hosting provider (single victim IP protected by Cloudflare).

Technology used: hyper-volumetric UDP flood with reflection/amplification; very high bandwidth (7.3 Tbps) delivered in a short burst (~45 seconds). Cloudflare measured ~37.4 TB of data delivered to one IP.

Attacker's motive (likely): disruptive testing of defenses, "proof-of-capability" or extortion, or to degrade a hosting provider (which in turn amplifies downstream outages). Public reporting did not attribute a named actor.

Overall impact: Cloudflare's network mitigated the attack automatically — the target would have likely suffered severe availability loss without a large-scale scrubbing service. The incident underscores the growing scale of volumetric attacks that can saturate ISP links. **Defensive strategies that could have mitigated it:**

- Always-on scrubbing via large scrubbing networks (Cloudflare, Gcore, Akamai) with unmetered capacity.
- Network-level egress/ingress filtering (BCP-38) to reduce spoofed traffic used in amplification.
- Rate-limiting and geo/IP blocks at upstream peering points; Anycast distribution to absorb volumetric load.
- Service-segmentation: don't expose management/control planes on public IPs; put critical infrastructure behind DDoS-hardened layers.
(These are the measures Cloudflare and others recommend for hyper-volumetric floods.

2) Cloudflare-mitigated 6.5 Tbps (April 2025) — near-record incident (impacted sites like KrebsOnSecurity)

Target: media/security blogs and a variety of hosting customers (high-profile targets included KrebsOnSecurity briefly).

Technology used: multi-vector network-layer floods (SYN/UDP floods, amplification via SSDP, Miraistyle botnet traffic) and extremely high packet rates (billions of pps reported for some events). **Attacker's motive (likely):** disruption of high-visibility sites (hacktivism), capability testing, or rentseeking extortion. Some campaigns appear to be sustained multi-day tests of DDoS capacity.

Overall impact: Cloudflare's autonomous systems absorbed and scrubbing prevented large outages for many customers; victims without cloud scrubbing protections would likely have been knocked offline. The incident also raised concerns about inter-ISP backbone saturation risks.

Defensive strategies:

- Contracting always-on DDoS protection with high capacity and automatic detection.
- Harden origin servers by hiding origin IPs behind CDNs and using WAF rules to drop obvious protocol abuse.

- Threat-intelligence sharing among ISPs to blackhole attack flows and coordinate upstream filtering.
- Implement application-level caching and graceful degradation to preserve core functionality under attack.

3) Microsoft Azure / Microsoft 365 outage — DDoS-triggered (30 July 2024)

Target: Microsoft Azure services and Microsoft 365 components (Azure portal, Azure App Services, some Microsoft 365 functionality).

Technology used: spike in traffic consistent with a distributed denial-of-service. Microsoft reported the initial cause as a DDoS and later explained that combination of the attack and an internal mitigation/configuration error amplified the outage's impact.

Attacker's motive (likely): unattributed — could be disruption, civil/hacktivist action, or opportunistic testing. Public reporting did not identify an actor.

Overall impact: widespread service disruption for hours (many customers experienced outages or degraded access); because Microsoft is a major cloud provider, downstream dependent services and customers were affected — demonstrating systemic risk when major cloud providers have availability issues.

Defensive strategies that could have mitigated it:

- Rigorous testing and validation of automated mitigation logic (avoid single points where a mitigation action can cascade into larger outages). Microsoft's own post-incident review emphasized this.
- Runbook drills and staged failover: ensure that mitigation changes can be rolled back quickly and safely.
- Multi-region redundancy and automated traffic re-routing to unaffected regions with pretested playbooks.
- Transparent, fast customer communications and customer-side resiliency (multi-cloud or hybrid setups) to limit service dependency.

4) Cloudflare — 5.6 Tbps Mirai-variant attack (Oct 29, 2024)

Target: an East Asian ISP / Magic Transit customer (Cloudflare reported the 5.6 Tbps incident in Q4-2024 reporting).

Technology used: Mirai-variant IoT botnet capable of UDP amplification; thousands of compromised IoT devices used to create a 5.6 Tbps flood (Cloudflare reported ~13,000 source IPs contributing).

Attacker's motive (likely): Mirai-style botnets are used for rent-a-DDoS services, capability demonstration, and disruption. The use of IoT devices traditionally indicates botnet monetization or testing.

Overall impact: the victim ISP would likely have experienced severe service degradation; Cloudflare's autonomous mitigation neutralized the flood but the event highlighted how even relatively small numbers of IoT devices can be aggregated to huge effect. **Defensive strategies:**

- Improve IoT hygiene (vendor and ISP responsibility): default credential removal, network segmentation, and forced updates.
- Upstream ingress filtering (drop spoofed packets) and cloud scrubbing for ISPs.
- Rapid sinkholing and collaboration with hosting/cloud providers to quarantine malicious sources.
- IoT device owners: isolate devices on separate VLANs, limit inbound access, and patch firmware.

5) Gcore — 6 Tbps DDoS (mid-October 2025 mitigation) — very recent

Target: large tech customers / multi-regional infrastructure (Gcore statement cites tech sector targets).

Technology used: UDP-dominant volumetric flood, high packet rate (reported ~5.3 Bpps in coverage) and likely mixed sourcing from compromised cloud instances and IoT devices; Gcore's report flagged geographic concentration in some sources.

Attacker's motive (likely): ability-testing, supply-chain/third-party weakening, or financially motivated DDoS-for-hire campaigns. Public statements emphasize the rise of professionalized DDoS operations.

Overall impact: Gcore's automated mitigation absorbed the attack; the episode reinforces that multiple providers (not just Cloudflare) are facing and stopping Tbps-scale attacks as they get more frequent. The ongoing trend increases pressure on mid-sized orgs that lack scrubbing contracts.

Defensive strategies:

- Use scrubbing services with high aggregate capacity and multi-provider redundancy (don't rely on a single vendor if you're critical infrastructure).
- ISP cooperation and blackholing at peering fabrics to protect global backbone links.
- Ensure cloud instances are not abused as unwitting amplifiers — cloud providers should monitor for abusive egress patterns and enforce stricter API/account controls.